Jungmin Kim, Ph.D.

Scientist I
Dept. of Electrical and Computer Engineering
University of Wisconsin-Madison, Madison, WI

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Education

Ph.D. Electrical and Computer Engineering — Mar 2017–Feb 2023
Seoul National University — Seoul, South Korea
Advisors: Prof. Namkyoo Park and Prof. Sunkyu Yu

B.S. Electrical and Computer Engineering — Mar 2012–Feb 2017
Seoul National University — Seoul, South Korea

Work Experience

University of Wisconsin–Madison — Madison, WI
Research Associate, Scientist I — Apr 2023–Present

- Mentor: Prof. Zongfu Yu
- Designed a mmWave face verification system using metasurfaces, achieving $> 90\%$ accuracy.
- Designed an incoherent meta-imaging system for low-light environments, with an approximately 2.9 dB PSNR enhancement.
- Designed a photonic systolic array for all-optical matrix–matrix multiplication, achieving a theoretical computing density of 4.4 PMACS$\cdot$mm^{-2}s^{-1}.

Seoul National University — Seoul, South Korea
Research / Teaching Assistant — Mar 2017–Feb 2023

- Designed tilted Dirac cones and investigated temporally disordered modulation in photonic systems to achieve target scattering responses.
- Assisted with ECE courses in Quantum Mechanics, Electromagnetism, and Nanophotonics.

Selected Projects

Optical computing — Related publications: [1–4, 6]

Using gradient-based optimization techniques, including the adjoint method and neural-network-based modeling of diffractive optical systems, I have inverse-designed various optical devices for neuromorphic and computing applications. These include neuromorphic metasurfaces for face recognition and output-stationary optical systolic arrays that accelerate matrix–matrix multiplication.

Disordered photonics — Related publications: [5, 7, 9]

Harnessing hidden degrees of freedom in material phase can unlock numerous functionalities. I used deep learning to inverse-design photonic multilayer structures with angle-dependent optical target responses and active tunability. I also extended the concept of correlated disorder—an intermediate regime between perfect order and complete disorder—into the temporal domain, enabling top-down control of light scattering through time-varying disordered modulation without requiring complex material structures.

Photonic band structure engineering Related publications: [8, 10]

Photonic Dirac cones enable exceptional light propagation with applications such as optical cloaking and light funneling, but their directional control has remained largely unexplored. To address this, I developed an inverse design approach that perturbs the eigenmodes of a conventional upright Dirac cone to create a tilted one, enabling more efficient and directional light transport in photonic devices.

Skills

EM simulations	FDTD (Tidy3D), FEM (COMSOL), RCWA, PWEM
Programming	MATLAB, Python
Optimization	Adjoint method, automatic differentiation, gradient-based inverse design
Machine learning	PyTorch

Publications

[1] **J. Kim**, Q. Zhou, and Z. Yu, "Photonic systolic array for all-optical matrix–matrix multiplication", Laser & Photonics Reviews **20**, e01995 (2026).

[2] Q. Zhou, **J. Kim**, Y. Tao, et al., "All-optical deep learning with quantum nonlinearity", arXiv, 2601.01690 (2026).

[3] **J. Kim**, N. Yu, and Z. Yu, "Compute-first optical detection for noise-resilient visual perception", ACS Photonics **12**, 1137–1145 (2025).

[4] J. Kim, J.-Y. Kim, **J. Kim**, et al., "Inverse design of nanophotonic devices enabled by optimization algorithms and deep learning: recent achievements and future prospects", Nanophotonics **14**, 121–151 (2025).

[5] **J. Kim**, D. Lee, S. Yu, and N. Park, "Unidirectional scattering with spatial homogeneity using correlated photonic time disorder", Nature Physics **19**, 726–732 (2023).

[6] S. Choi, **J. Kim**, J. Kwak, N. Park, and S. Yu, "Topologically protected all-optical memory", Advanced Electronic Materials **8**, 2200579 (2022).

[7] S. Oh, **J. Kim**, X. Piao, et al., "Control of localization and optical properties with deep-subwavelength engineered disorder", Optics Express **30**, 28301–28311 (2022).

[8] S. Park, I. Lee, **J. Kim**, N. Park, and S. Yu, "Hearing the shape of a drum for light: isospectrality in photonics", Nanophotonics **11**, 2763–2778 (2022).

[9] **J. Kim**, S. Park, S. Yu, and N. Park, "Machine-engineered active disorder for digital photonics", Advanced Optical Materials **10**, 2102642 (2022).

[10] **J. Kim**, S. Yu, and N. Park, "Universal design platform for an extended class of photonic dirac cones", Physical Review Applied **13**, 044015 (2020).

Presentations

[C1] J. Kim, N. Yu, and Z. Yu, "Incoherent meta-imaging system for noise-robust object recognition", in 2024 conference on lasers and electro-optics pacific rim (CLEO-PR) (2024), Mo1A4.

[C2] D. Lee, J. Kim, H. Park, et al., “Design of correlated photonic time disorder for unidirectional scattering”, in Advanced photonics congress 2023 (2023), NoTu3C.4.

[C3] J. Kim, S. Park, D. Lee, S. Yu, and N. Park, “Data-driven engineering of active photonic disorder”, in Frontiers in optics + laser science 2022 (FIO+LS) (2022), JW4A.20.

[C4] D. Lee, J. Kim, N. Park, and S. Yu, “Molecular dynamics for microscopic analysis of refractive index in amorphous hafnium oxides”, in Frontiers in optics + laser science 2022 (FIO+LS) (2022), FW7C.5.

[C5] J. Kim, S. Yu, and N. Park, “Neural-network-based design of tunable multilayer films”, in OSA advanced photonics congress 2021 (2021), JW4B.3.

Honors and Awards

Feb 2023	Distinguished Dissertation Award, ECE, Seoul National University
Feb 2019	Best Poster Award, OSK Winter Meeting
2014–2016	National Science & Engineering Undergraduate Scholarship, Korea Student Aid Foundation

Academic Service

Reviewer	Optica, Photonics Research, Optics Express, Optics Letters Nature Electronics, Nature Communications ACS Photonics, Advanced Functional Materials

Last updated on April 8, 2026